

Programming in C# Lab 1

Objectives:

At the end of this session, you will be able to understand:

- ❖ Data type in C#
- ❖ Input and output data in C#
- ❖ Number and datetime format specifier

Part I: Workshop – 15 minutes

Students open workshop in CD ROM, then View, Run, Think about it.

Part II: Step by step – 45 minutes

Exercise 1: Display a message

Step 1: Open Visual Studio 2005

Step 2: Select the menu File->New->Project to create console based project named 'First_prg' and Solution named Session01 as shown in Figure 1

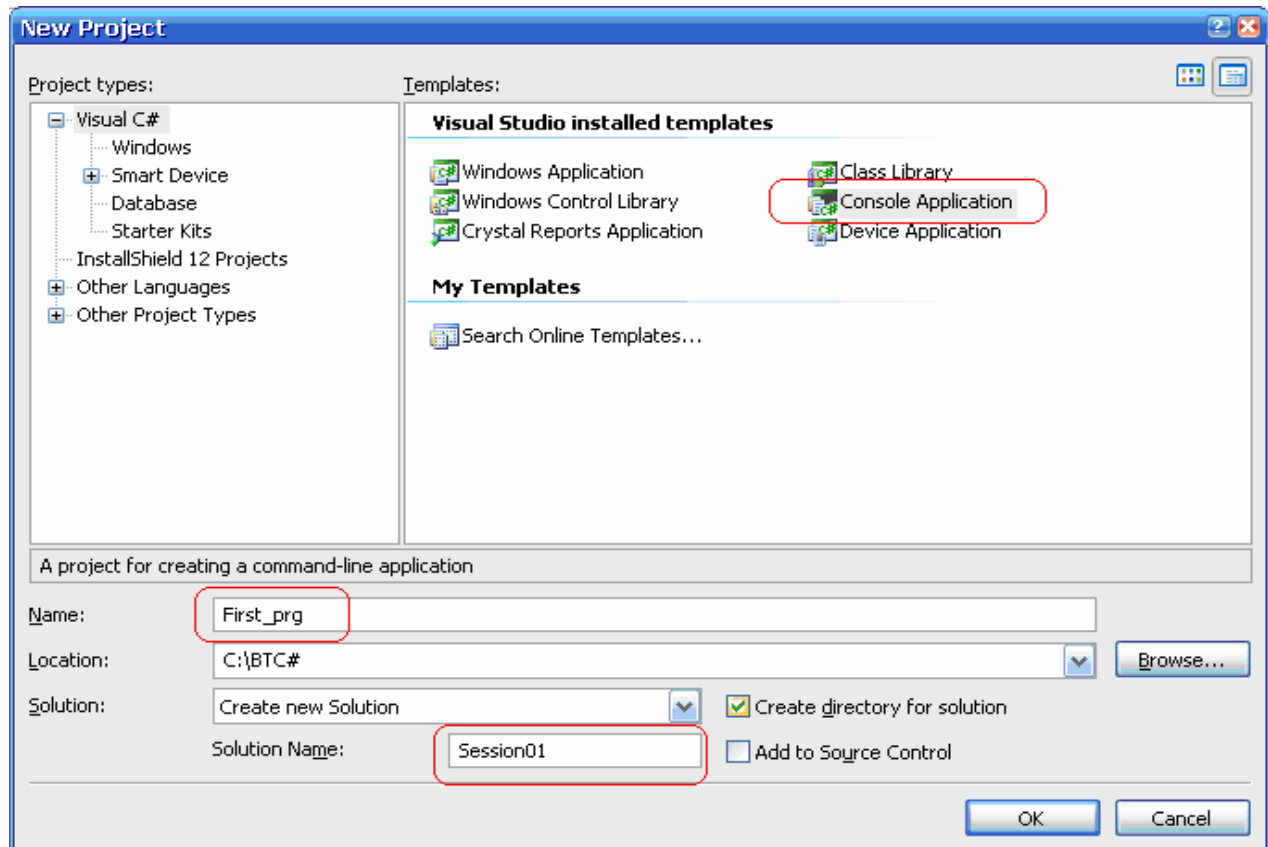


Figure 1: New Project Dialog Box

Step 3: Rename the class file 'program.cs' to 'First_prg.cs'

Step 4: Replace code in 'First_prg.cs' with given code

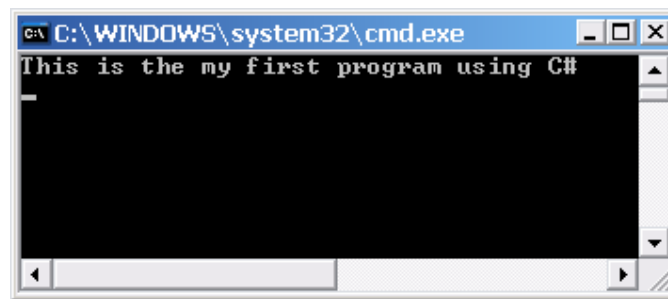
```
using System;
class Example
{
    static void Main(string[] args)
    {
        Console.WriteLine("This is the my first program using C#");
        Console.ReadLine();
    }
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build First_prg option to build the project

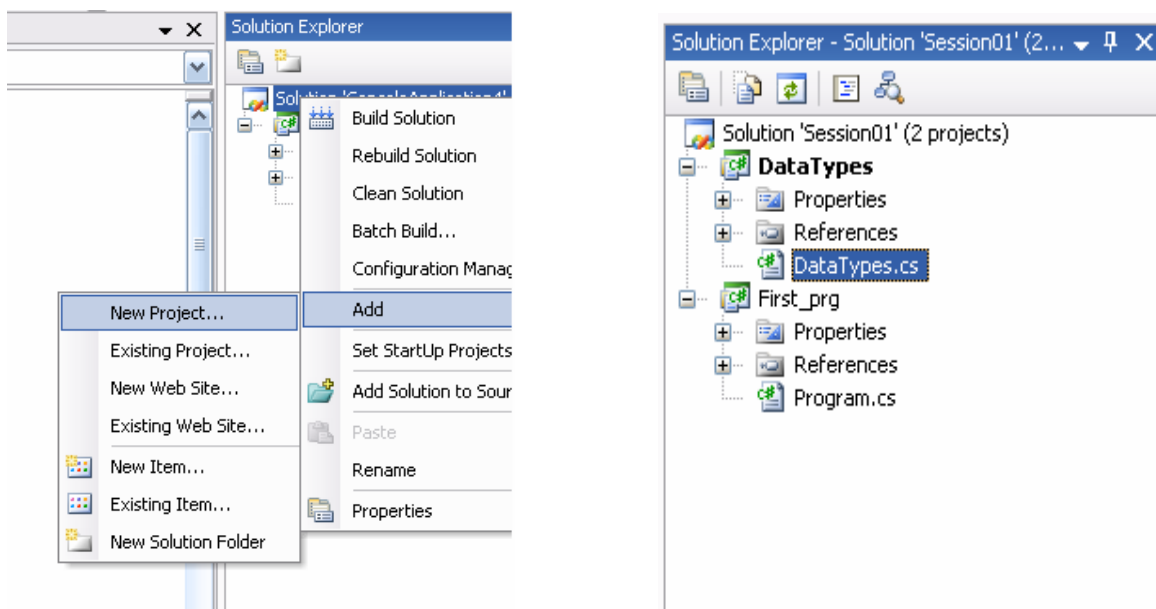
Step 7: Select Debug -> Start without Debugging to execute the program

The output of the program as following



Exercise 2: Using datatype

Step 1: Add a console based project 'DataType' to the solution



Step 2: Right click on project DataTypes -> set as Startup project

Step 3: Rename the class file 'Program.cs' to 'DataTypes.cs'

Step 4: Replace the code in 'DataTypes.cs' with the given code

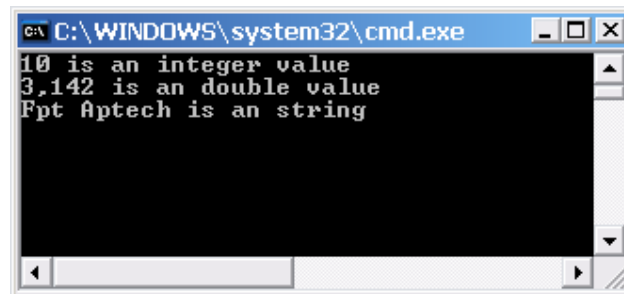
```
using System;
class Example
{
    static void Main(string[] args)
    {
        int intVal;
        double dblVal;
        string strVal;
        intVal = 10;
        dblVal = 3.142;
        strVal = "Fpt Aptech";
        Console.WriteLine("{0} is an integer value", intVal);
        Console.WriteLine("{0} is an double value", dblVal);
        Console.WriteLine("{0} is an string", strVal);
        Console.ReadLine();
    }
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build 'DataTypes' option to build the project

Step 7: Select Debug -> Start without Debugging to execute the program

The output of program as following



Exercise 3: value type

Step 1: Add a console based project 'ValueType' to the solution

Step 2: Right click on project ValueType-> set as Startup project

Step 3: Rename the class file 'Program.cs' to 'ValueType.cs'

Step 4: Replace the code in 'ValueType.cs' with the given code

```
using System;
using System.Text;

class Example1
```

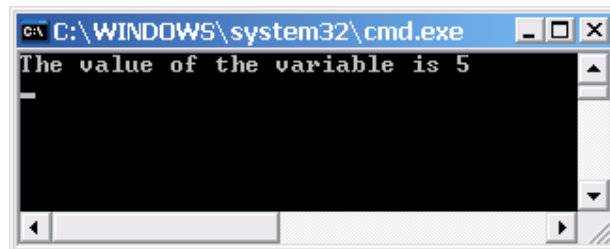
```
{
    static void Main(string[] args)
    {
        int valueVal = 5;
        Test(valueVal);
        Console.WriteLine("The value of the variable is {0}", valueVal);
        Console.ReadLine();
    }
    static void Test(int valueVal)
    {
        int temp = 5;
        valueVal = temp * 2;
    }
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build 'ValueType' option to build the project

Step 7: Select Debug -> Start without Debugging to execute the program

The output of program as following



Exercise 3: Reference Type

Step 1: Add a console based project 'ReferenceType' to the solution

Step 2: Right click on project ReferenceType -> set as Startup project

Step 3: Rename the class file 'Program.cs' to 'ReferenceType.cs'

Step 4: Replace the code in 'ReferenceType.cs' with the given code

```
using System;
class ReferenceType
{
    public int valueVal;
}
class TestReference
{
    static void Main(string[] args)
    {
        ReferenceType refer = new ReferenceType();
    }
}
```

C# - Lab 1 – Data Type, Input and Output, Conversion, format specifier

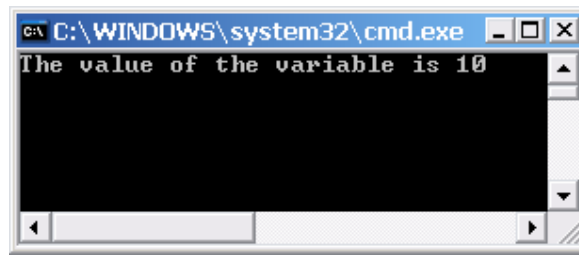
```
        refer.valueVal = 5;
        Test(refer);
        Console.WriteLine("The value of the variable is {0}",
refer.valueVal);
        Console.ReadLine();
    }
    static void Test(ReferenceType refer)
    {
        int temp = 5;
        refer.valueVal = temp * 2;
    }
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build ReferenceType option to build the project

Step 7: Select Debug -> Start without Debugging to execute the program

The output of program as following



Exercise 4: output

Step 1: Add a console based project 'Output' to the solution

Step 2: Right click on project Output -> set as Startup project

Step 3: Rename the class file 'Program.cs' to 'Output.cs'

Step 4: Replace the code in 'Output.cs' with the given code

```
/* This program demonstrates the output operators in C# */
using System;
class Student
{
    static void Main(string[] args)
    {
        //Declaring and intialising variables to store student details
        int id = 1;
        string name = "David George";
        byte age = 18;
        char gender = 'M';
        float percent = 75.50F;
    }
}
```

C# - Lab 1 – Data Type, Input and Output, Conversion, format specifier

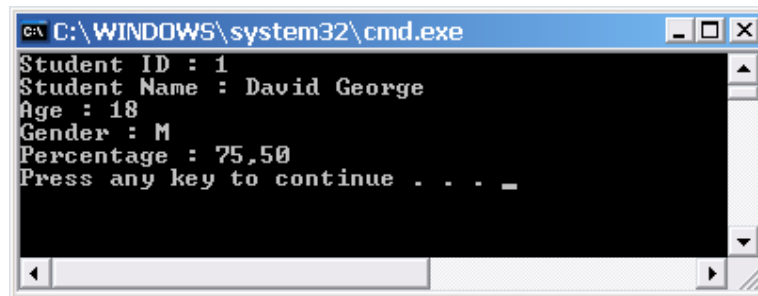
```
//Displaying the student details
Console.WriteLine("Student ID : {0}", id);
Console.WriteLine("Student Name : {0}", name);
Console.WriteLine("Age : " + age);
Console.WriteLine("Gender : " + gender);
Console.WriteLine("Percentage : {0:F2}", percent);
    }
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build Output option to build the project

Step 7: Select Debug -> Start without Debugging to execute the program

The output of program as following



```
C:\WINDOWS\system32\cmd.exe
Student ID : 1
Student Name : David George
Age : 18
Gender : M
Percentage : 75,50
Press any key to continue . . .
```

Exercise 5: Input

Step 1: Add a console based project 'Input' to the solution

Step 2: Right click on project Input -> set as Startup project

Step 3: Rename the class file 'Program.cs' to 'Input.cs'

Step 4: Replace the code in 'Input.cs' with the given code

```
using System;
/* The program demonstrates the input and output operations.*/
class Student
{
    static void Main(string[] args)
    {
        //Declaring integer constant to store value 100
        const int percentConst = 100;
        //Declaring variable to store the student name
        string studentName;

        //Declaring variables to store the student marks
        int english, maths, science;
        //Declaring and initialising variable to store the percentage
```

```
float percent = 0.0F;
//Accepting the details of the student
Console.WriteLine("Enter name of the student : ");
studentName = Console.ReadLine();

Console.WriteLine("Enter marks for english : ");
english = Convert.ToInt32(Console.ReadLine());

Console.WriteLine("Enter marks for maths : ");
maths = Convert.ToInt32(Console.ReadLine());

Console.WriteLine("Enter marks for science : ");
science = Convert.ToInt32(Console.ReadLine());

//Calculating the percentage of the student
percent = ((english + maths + science) * percentConst) / 300;

//Displaying the details of the student
Console.WriteLine("Student Name : " + studentName);

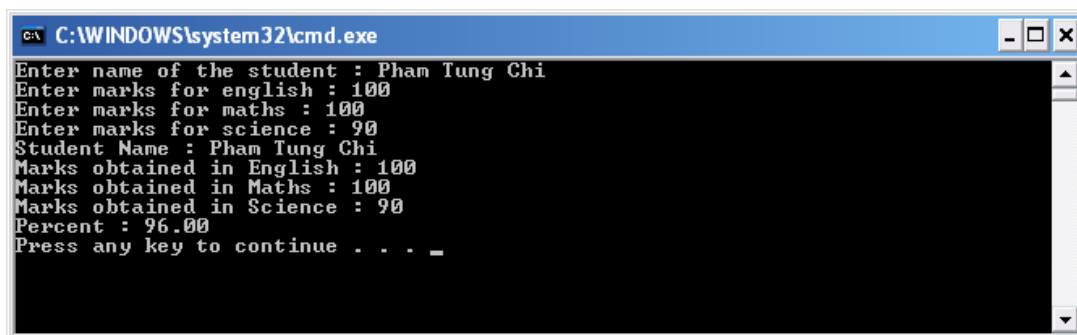
Console.WriteLine("Marks obtained in English : {0}", english);
Console.WriteLine("Marks obtained in Maths : {0}", maths);
Console.WriteLine("Marks obtained in Science : {0}", science);
Console.WriteLine("Percent : {0:F2}", percent);
}
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build 'Input' option to build the project

Step 7: Select Debug -> Start without Debugging to execute the program

The output of program as following



```
C:\WINDOWS\system32\cmd.exe
Enter name of the student : Pham Tung Chi
Enter marks for english : 100
Enter marks for maths : 100
Enter marks for science : 90
Student Name : Pham Tung Chi
Marks obtained in English : 100
Marks obtained in Maths : 100
Marks obtained in Science : 90
Percent : 96.00
Press any key to continue . . . _
```

Exercise 5: Number format specifier

Step 1: Add a console based project 'NumberFormat' to the solution

C# - Lab 1 – Data Type, Input and Output, Conversion, format specifier

Step 2: Right click on project NumberFormat -> set as Startup project

Step 3: Rename the class file 'Program.cs' to 'NumberFormat.cs'

Step 4: Replace the code in 'NumberFormat.cs' with the given code

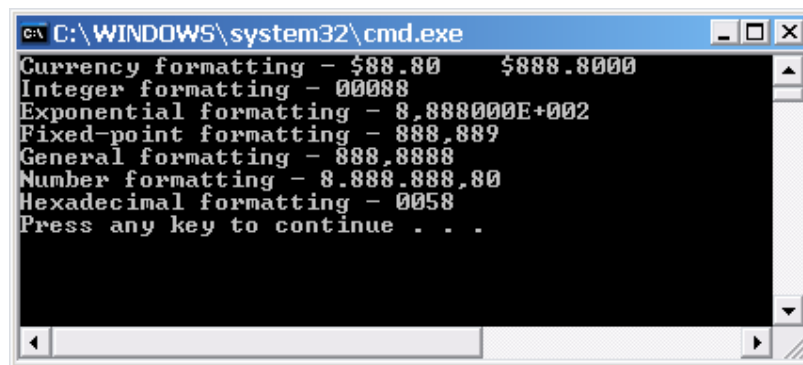
```
/*This program demonstrates the the numeric formatting in C#*/  
using System;  
class NumberFormat  
{  
    static void Main(string[] args)  
    {  
        Console.WriteLine("Currency formatting - {0:C}      {1:C4}", 88.8, 888.8);  
        Console.WriteLine("Integer formatting - {0:D5}", 88);  
        Console.WriteLine("Exponential formatting - {0:E}", 888.8);  
        Console.WriteLine("Fixed-point formatting - {0:F3}", 888.8888);  
        Console.WriteLine("General formatting - {0:G}", 888.8888);  
        Console.WriteLine("Number formatting - {0:N}", 8888888.8);  
        Console.WriteLine("Hexadecimal formatting - {0:X4}", 88);  
    }  
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build 'NumberFormat' option to build the project

Step 7: Select Debug -> Start without Debugging to execute the program

The output of program as following



The screenshot shows a Windows command prompt window titled "C:\WINDOWS\system32\cmd.exe". The output of the program is as follows:

```
Currency formatting - $88.80      $888.8000  
Integer formatting - 00088  
Exponential formatting - 8.888000E+002  
Fixed-point formatting - 888.889  
General formatting - 888.8888  
Number formatting - 8.888.888.80  
Hexadecimal formatting - 0058  
Press any key to continue . . .
```

Exercise 5: Datetime format specifier

Step 1: Add a console based project 'DateTimeFormat' to the solution

Step 2: Right click on project DateTimeFormat -> set as Startup project

Step 3: Rename the class file 'Program.cs' to 'DateTimeFormat.cs'

Step 4: Replace the code in 'DateTimeFormat.cs' with the given code

```
using System;
class MainClass
{
    public static void Main()
    {
        DateTime dt = DateTime.Now; // obtain current time

        Console.WriteLine("d format: {0:d}", dt);
        Console.WriteLine("D format: {0:D}", dt);

        Console.WriteLine("t format: {0:t}", dt);
        Console.WriteLine("T format: {0:T}", dt);

        Console.WriteLine("f format: {0:f}", dt);
        Console.WriteLine("F format: {0:F}", dt);

        Console.WriteLine("g format: {0:g}", dt);
        Console.WriteLine("G format: {0:G}", dt);

        Console.WriteLine("m format: {0:m}", dt);
        Console.WriteLine("M format: {0:M}", dt);

        Console.WriteLine("r format: {0:r}", dt);
        Console.WriteLine("R format: {0:R}", dt);

        Console.WriteLine("s format: {0:s}", dt);

        Console.WriteLine("u format: {0:u}", dt);
        Console.WriteLine("U format: {0:U}", dt);

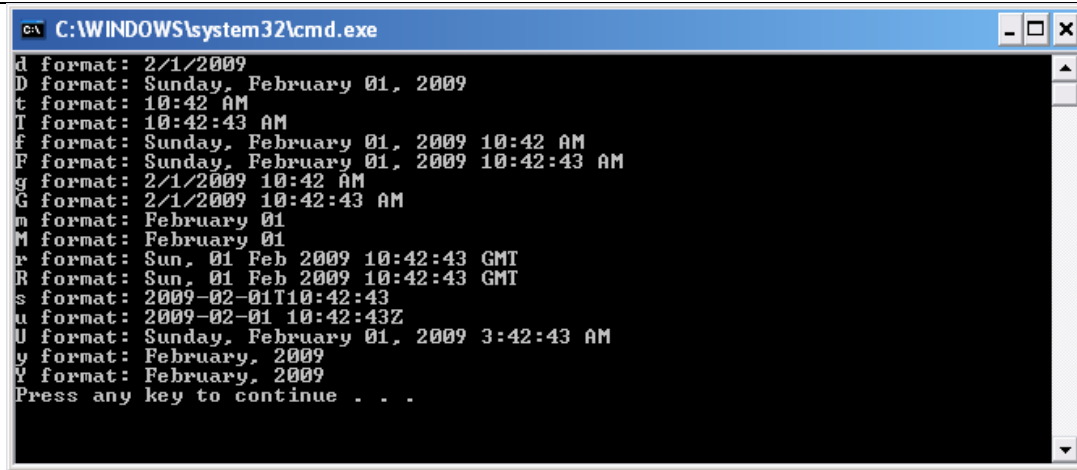
        Console.WriteLine("y format: {0:y}", dt);
        Console.WriteLine("Y format: {0:Y}", dt);
    }
}
```

Step 5: Select menu File -> Save to save the file

Step 6: Select Build -> Build 'DateTimeFormat' option to build the project

Step 7: Select Debug -> Start without Debugging to execute the program

The output of program as following



```
C:\WINDOWS\system32\cmd.exe
d format: 2/1/2009
D format: Sunday, February 01, 2009
t format: 10:42 AM
T format: 10:42:43 AM
f format: Sunday, February 01, 2009 10:42 AM
F format: Sunday, February 01, 2009 10:42:43 AM
g format: 2/1/2009 10:42 AM
G format: 2/1/2009 10:42:43 AM
m format: February 01
M format: February 01
r format: Sun, 01 Feb 2009 10:42:43 GMT
R format: Sun, 01 Feb 2009 10:42:43 GMT
s format: 2009-02-01T10:42:43
u format: 2009-02-01 10:42:43Z
U format: Sunday, February 01, 2009 3:42:43 AM
y format: February, 2009
Y format: February, 2009
Press any key to continue . . .
```

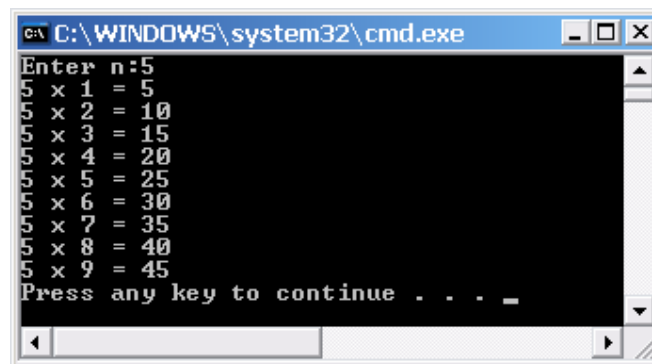
Part III: Do it yourself – 60 minutes

Exercise 1: Write a program to enter: name, address, phone and display these information.

Exercise 2: Write a program to accept three integer number and find maximum number from three integer.

Exercise 3: Write a program that accepts a number between 1 and 7 from the user and return the corresponding day of the week(1- Monday, 2- Tuesday and so on).

Exercise 4: Write a program to display the first 9 multiples of an integer. N entered from user



```
C:\WINDOWS\system32\cmd.exe
Enter n:5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
Press any key to continue . . .
```

Exercise 5: Write a program to print the factorials of the integers from 1 to 20

Part IV: Homework

Exercise 1: Do assignment of module 1 in CD ROM

Exercise 2: Do assignment of module 2 in CD ROM

Exercise 3: Do assignment of module 3 in CD ROM

References

- 1) CD ROM C# Programming, Aptech Education
- 2) <http://www.java2s.com/Tutorial/CSharp/CatalogCSharp.htm>
- 3) MSDN Document